

AN FTP

FTP Service — Section 1: Service Brief

1.1 What is FTP?

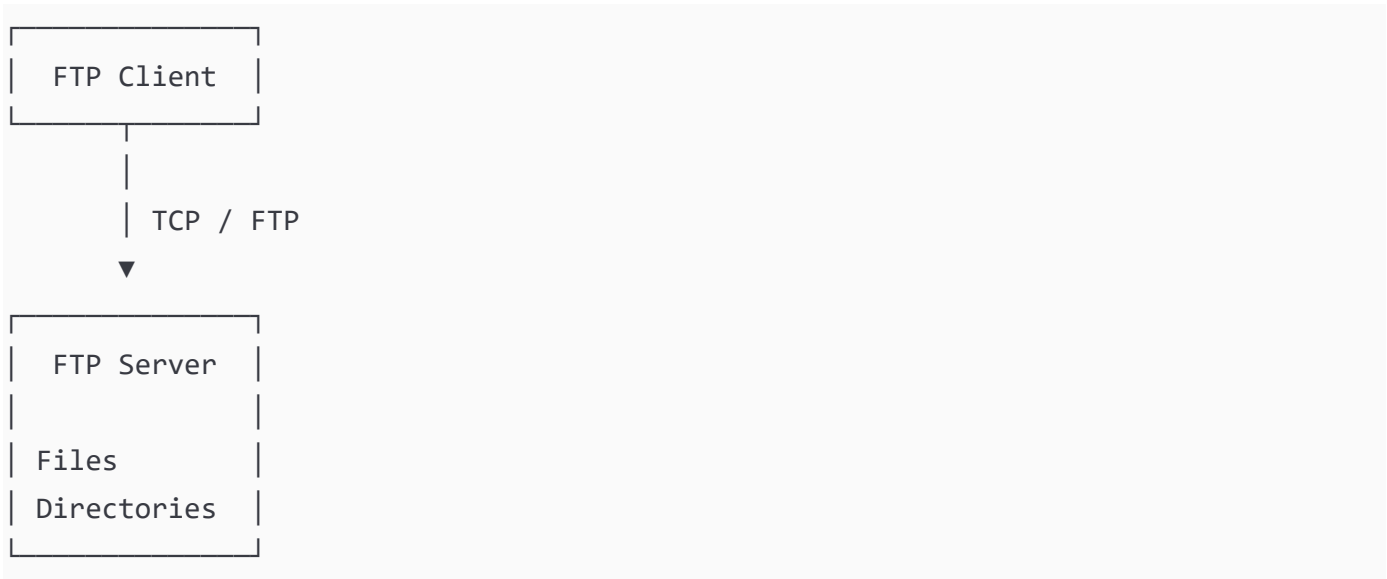
FTP (File Transfer Protocol) is a **TCP-based client-server protocol** used to transfer files between computers over a network.

An **FTP client** connects to an **FTP server** to upload, download, and manage files and directories.

1.2 Key Facts

Feature	FTP
Full Name	File Transfer Protocol
Architecture	Client–Server
Transport	TCP
Control Port	21
File Transfer	Upload / Download
Modes	Active / Passive
Encryption	❌ Not provided by plain FTP

1.3 Basic FTP Architecture



FTP uses a **control connection** for commands and a separate **data connection** for file or directory data. The data connection operates differently in **Active and Passive modes**.

1.4 Security Note

Plain FTP does not encrypt communication. User credentials and transferred data can therefore be exposed on an untrusted network.

For encrypted file transfer:

- **FTPS** → FTP secured with TLS
 - **SFTP** → File transfer through SSH
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1.5 Section Summary

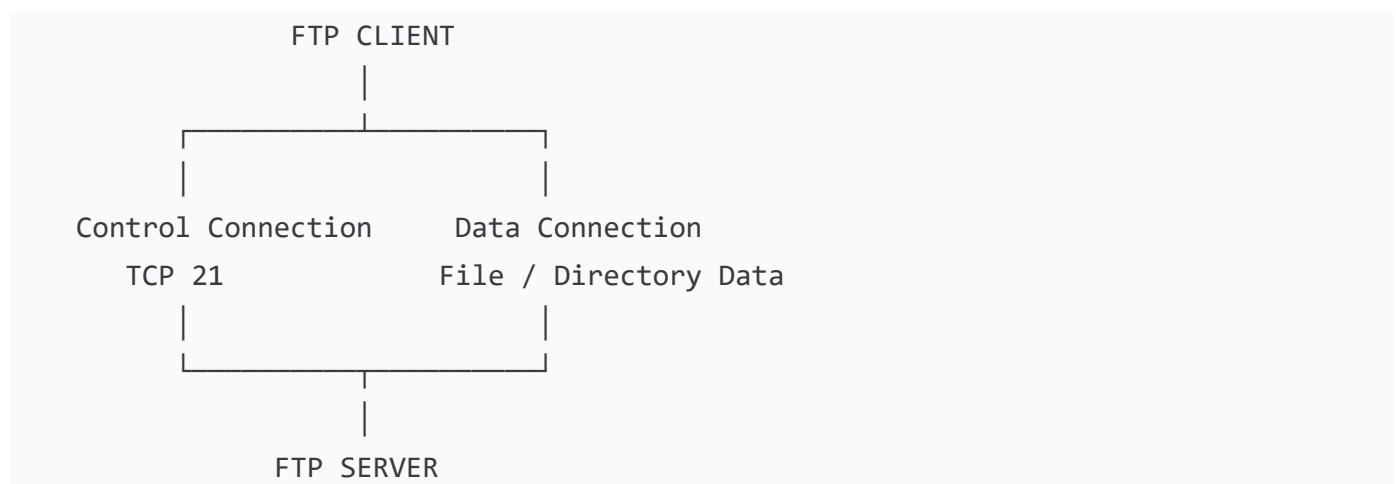
FTP is a TCP-based client-server protocol used for network file transfer. Its separate control and data connections make Active and Passive modes important concepts for FTP administration.

Section 2: Architecture & Working

2.1 FTP Communication Model

FTP uses a **client-server architecture** and establishes two separate types of TCP connections:

1. **Control Connection** — carries commands and server responses.
2. **Data Connection** — carries files and directory listings.



The control connection normally remains established during the FTP session, while data connections are created as required for data transfers.

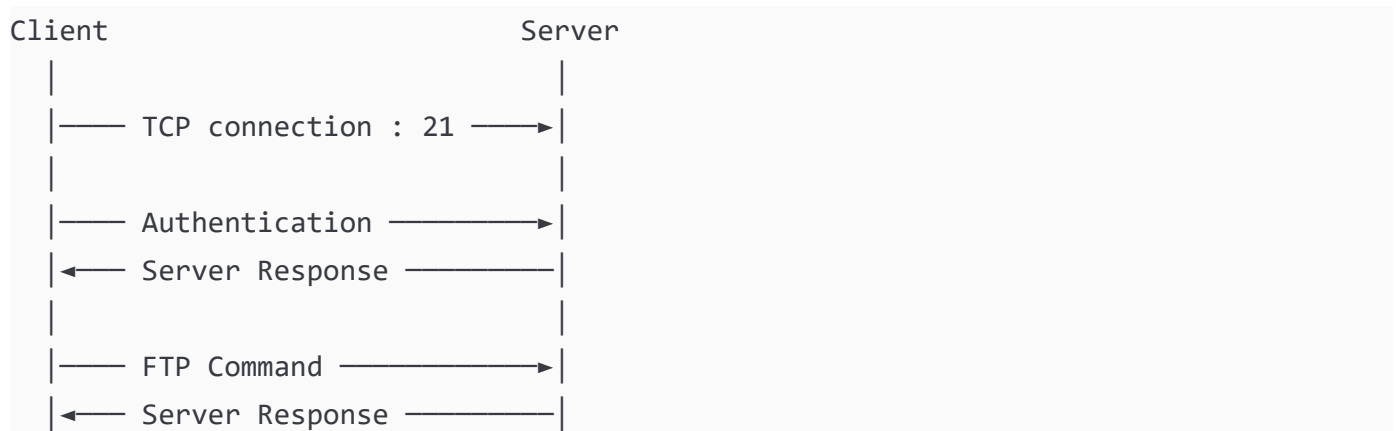
2.2 Control Connection

The **control connection** is normally established between the client and the server on **TCP port 21**.

It is used for:

- User authentication
- FTP commands
- Directory navigation
- Transfer requests
- Server responses

Example:



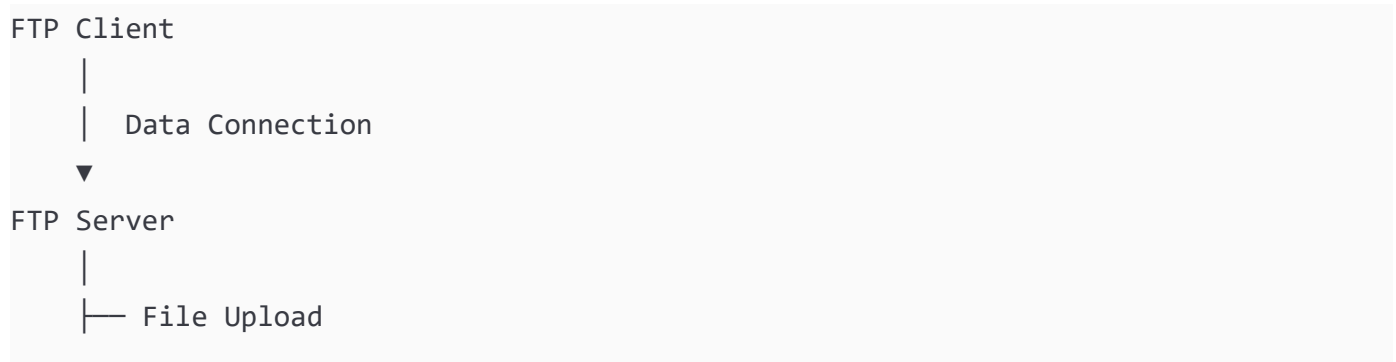
The actual file contents are **not normally transferred through the control connection**.

2.3 Data Connection

The **data connection** carries the actual data requested by the client.

It is used for:

- File uploads
- File downloads
- Directory listings

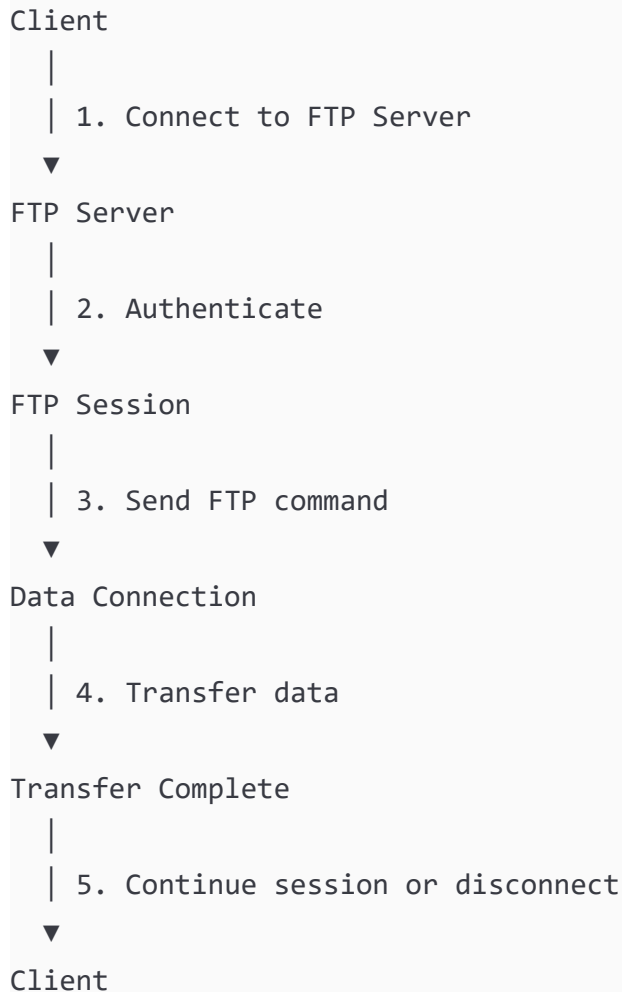


- └ File Download
- └ Directory Listing

Unlike the control connection, the data connection is created **when data needs to be transferred**.

2.4 FTP Session Flow

A simplified FTP session follows this sequence:



2.5 Why Does FTP Need Two Connections?

FTP separates **control communication** from **data transfer**.

This allows commands and server responses to use the control connection while files and directory listings are transferred through a separate data connection.

Therefore:

TCP 21 handles FTP control communication, while the data connection handles the actual file or directory data.

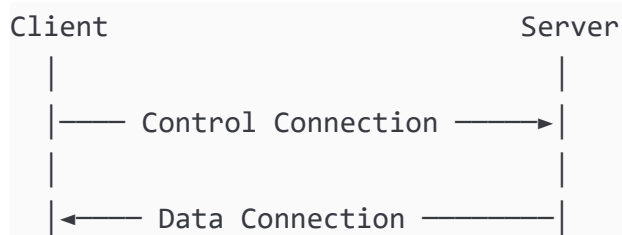
The way this separate data connection is established is covered in **Active and Passive FTP**.

FTP uses two separate connections: one for control and one for data.

Section 3: Active & Passive FTP

3.1 Active FTP

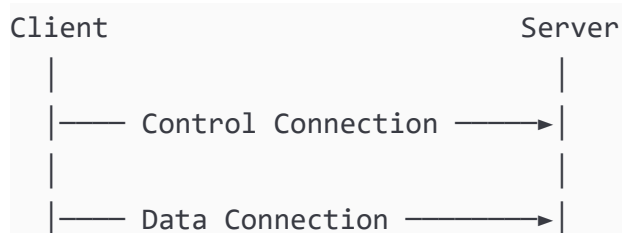
In **Active FTP**, the client establishes the control connection, while the **server initiates the data connection back to the client**.



- Control connection → TCP **21**
 - Data connection → traditionally TCP **20** on the server side
 - Server initiates the data connection.
 - More difficult to use through **firewalls and NAT** because the server connects back to the client.
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3.2 Passive FTP

In **Passive FTP**, the client establishes **both the control and data connections** with the server.



- Control connection → TCP **21**
 - Server provides a temporary **passive data port**.
 - Client initiates the data connection to that port.
 - The server normally uses a configured **passive port range**.
 - Generally easier to use through **firewalls and NAT**.
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3.3 Active vs Passive FTP

Feature	Active FTP	Passive FTP
Control connection	Client → Server	Client → Server
Data connection	Server → Client	Client → Server
Data port	Traditionally TCP 20	Configured passive port range
Firewall/NAT	More difficult	Generally easier
Common deployment	Less preferred	Commonly preferred

3.4 Important Point

Active and Passive FTP **are not different FTP services**. They are two different methods of establishing the **FTP data connection**.

Active → Server initiates the data connection.

Passive → Client initiates the data connection.

The actual **passive port configuration, firewall rules, and testing** will be covered in the configuration and practical implementation sections.

Section 4: Prerequisites & Dependencies

4.1 Lab Requirements

For the practical setup:

Machine	Role
Machine 1	FTP Server — Debian/Linux with <code>vsftpd</code>
Machine 2	FTP Client — Debian/Linux with FTP client

Both machines must have **network connectivity** between them.

4.2 Basic Requirements

FTP Server

- Debian/Linux system
- Reachable IP address
- `vsftpd` package
- Local user for FTP access

- Required storage directory

FTP Client

- Debian/Linux system
- FTP client software
- Network access to the FTP server
- Valid FTP credentials

4.3 Key Dependencies

Network

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FTP Server (vsftpd)

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├─ User Authentication

└─ File Permissions

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FTP Client (ftp)

The FTP service depends mainly on network connectivity, user authentication, and filesystem permissions.

4.4 Pre-Installation Check

- ☐ Server and client are ready
- ☐ Server IP address is known
- ☐ Client can reach the server
- ☐ FTP user is available

Section 5: Installation

5.1 Install vsftpd

On the **FTP Server**:

```
sudo apt update
```

```
sudo apt install vsftpd
```

5.2 Start and Enable Service

```
sudo systemctl start vsftpd
sudo systemctl enable vsftpd
```

Check the service:

```
sudo systemctl status vsftpd
```

5.3 Verify FTP Port

```
sudo ss -lntp | grep :21
```

Section 6: FTP Server Configuration

6.1 Main Configuration File

The main `vsftpd` configuration file is:

```
/etc/vsftpd.conf
```

create a backup:

```
sudo cp /etc/vsftpd.conf /etc/vsftpd.conf.bak
```

6.2 Basic Configuration

Edit the configuration file:

```
sudo nano /etc/vsftpd.conf
```

For a basic authenticated FTP server, ensure:

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
```

Parameter	Purpose
<code>anonymous_enable=NO</code>	Disable anonymous login
<code>local_enable=YES</code>	Allow local Linux users
<code>write_enable=YES</code>	Allow write operations such as upload

Important: `write_enable=YES` enables write operations on the **FTP server**. It does not mean "two-way connection" and does not need to be configured on the client.

6.3 Apply Configuration

After modifying the configuration:

```
sudo systemctl restart vsftpd
```

Check the service:

```
sudo systemctl status vsftpd
```

Section 7: Practical FTP Setup

7.1 Create FTP User

On the **FTP Server**, create a local Linux user:

```
sudo adduser ftpuser
```

`adduser` automatically creates the user's home directory.

7.2 Create a Test File

Create one test file in the user's home directory:

```
sudo touch /home/ftpuser/test.txt
```

7.3 Restart FTP Service

After the configuration and user setup:

```
sudo systemctl restart vsftpd
```

7.4 Connect from FTP Client

From the **FTP Client**, connect to the FTP server using:

```
ftp ftpuser@SERVER_IP
```

For example:

```
ftp ftpuser@192.168.1.10
```

Enter the password of `ftpuser` when prompted.

Note: `ftp username@IP` identifies the FTP user and server. It does not enable two-way communication. The FTP server configuration controls what operations the user is allowed to perform.

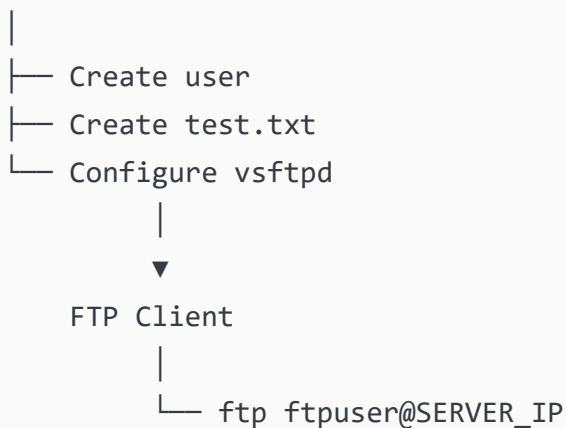
7.5 Basic FTP Commands

After successful login:

Command	Purpose
<code>ls</code>	List remote files
<code>pwd</code>	Show remote directory
<code>cd</code>	Change remote directory
<code>get</code>	Download a file
<code>put</code>	Upload a file
<code>bye</code> / <code>quit</code>	Exit FTP

Practical Flow

FTP Server



Section 8: FTP Client Operations

8.1 Navigate Remote Files

After connecting to the FTP server:

```
pwd
ls
cd directory_name
```

- `pwd` → Shows the current remote directory.

- `ls` → Lists files and directories.
 - `cd` → Changes the remote directory.
-

8.2 Download Files

Use `get` to download a file from the server:

```
get test.txt
```

The file is downloaded to the client's current local directory.

8.3 Upload Files

Use `put` to upload a file to the server:

```
put test.txt
```

The file is transferred from the client to the FTP server.

8.4 Disconnect

Use either:

```
bye
```

to close the FTP session.

Section 9: Security

9.1 Disable Anonymous Access

For an authenticated FTP server, disable anonymous login:

```
anonymous_enable=NO
```

This ensures that users must authenticate before accessing the FTP service.

9.2 Use Local User Authentication

Allow only required Linux users:

```
local_enable=YES
```

Access should be given only to users who need FTP service.

9.3 Protect FTP Credentials

Plain FTP does **not encrypt** usernames, passwords, or file contents.

Therefore:

- Avoid plain FTP for sensitive data.
 - Use **FTPS** when FTP with TLS encryption is required.
 - Use **SFTP** for SSH-based secure file transfer.
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Section 10: Troubleshooting

10.1 Check Service

```
sudo systemctl status vsftpd
```

If required:

```
sudo systemctl restart vsftpd
```

10.2 Check FTP Port

```
sudo ss -lntp | grep :21
```

If TCP **21** is listening, the FTP control service is available.

10.3 Login Failure

If the FTP login fails, check:

- Username and password
 - User account exists
 - `local_enable=YES`
 - `vsftpd` service status
-

10.4 Upload Failure

If login works but file upload fails, check:

- `write_enable=YES`

- Linux file/directory permissions
 - User ownership
-

10.5 File Download Failure

If downloading fails, check:

- File exists on the server
 - User has read permission
 - FTP data connection
 - Firewall rules
-

10.6 Directory Listing or Data Transfer Failure

If `ls`, `get`, or `put` fails while login works, check:

- Active/Passive FTP mode
 - Firewall configuration
 - Passive port range when using Passive FTP
 - Network connectivity
-

10.7 Check Logs

For more detailed information, check the system logs:

```
sudo journalctl -u vsftpd
```

Troubleshooting Flow

FTP Problem

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Service → Port → Login → Permissions

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Data Connection

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Firewall

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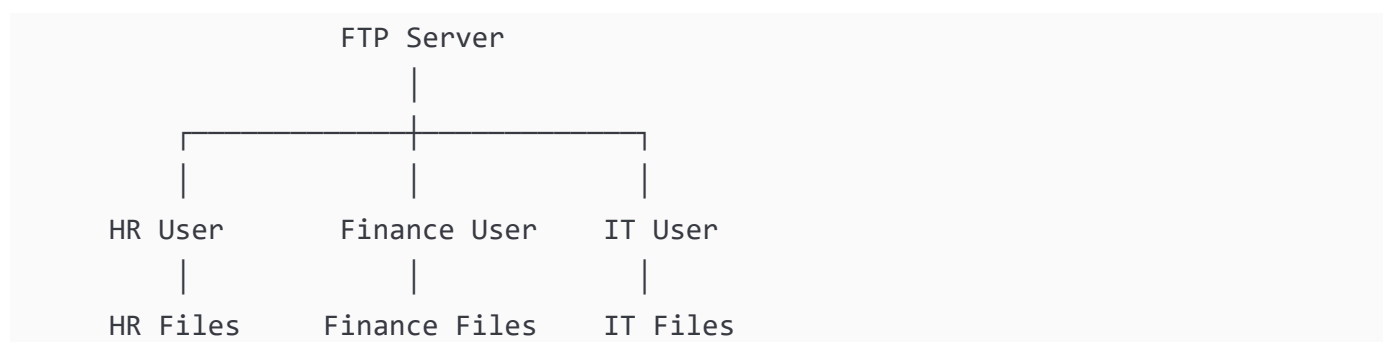
Logs

Always troubleshoot from the basic service and network checks toward authentication, permissions, data connections, and firewall issues.

Section 11: Real-World Scenario

11.1 Department-Based File Access

A company has multiple departments using the same FTP server. Each department has its own FTP user and directory.



For example:

- **HR user** → `/home/hr/`
- **Finance user** → `/home/finance/`
- **IT user** → `/home/it/`

Each user should normally have access only to the files required for their department.

11.2 Cross-Department File Requirement

Suppose the **HR department** needs an important report stored in the **Finance department's directory**.



The HR user should **not automatically get access to the entire Finance directory**.

Instead, the administrator can provide access to only the required file or a specific shared directory, according to the organization's permission policy.

11.3 Administrative Consideration

This scenario demonstrates why FTP administration requires:

- Separate users for departments
- Proper directory permissions
- Controlled cross-department access
- Restricted access to sensitive files
- Appropriate ownership and permissions

Practical goal: Allow departments to use the same FTP server while ensuring that users can access only the files and directories they are authorized to use.

Section 12: Verification Checklist

- ☐ `vsftpd` service is running.
- ☐ TCP **21** is listening.
- ☐ FTP client can connect and authenticate.
- ☐ Directory listing works.
- ☐ File download works.
- ☐ File upload works when permitted.
- ☐ User permissions restrict unauthorized access.
- ☐ Passive FTP works if configured.

Section 13: Viva / Interview Questions

Basic

1. What is FTP?

FTP is a TCP-based protocol used to transfer files between a client and server.

2. Which port does FTP use for the control connection?

TCP 21.

3. Why does FTP use two connections?

One for **control communication** and another for **data transfer**.

4. What is Active FTP?

The server initiates the data connection toward the client.

5. What is Passive FTP?

The client initiates the data connection to a server-provided passive port.

Administration

6. What is `vsftpd`?

A commonly used FTP server software for Linux.

7. Where is the main `vsftpd` configuration file?

`/etc/vsftpd.conf`

8. What does `write_enable=YES` do?

It enables FTP write operations such as uploading files.

9. Why is Passive FTP commonly preferred in modern networks?

It generally works more easily with **firewalls and NAT**.

10. Is plain FTP secure for sensitive data?

No. Plain FTP does not encrypt credentials or file contents.

Troubleshooting

11. Login works, but file transfer fails. What would you check?

Check the **data connection, Active/Passive mode, firewall, passive ports, and permissions**.

12. How do you check whether `vsftpd` is running?

```
sudo systemctl status vsftpd
```

13. How do you check whether FTP is listening on TCP 21?

```
sudo ss -lntp | grep :21
```

Key viva point: The most important FTP concepts are **control vs data connections, Active vs Passive mode, authentication, permissions, and FTP security**.

Section 14: Quick Reference

14.1 Important Paths

Path	Purpose
<code>/etc/vsftpd.conf</code>	Main FTP configuration
<code>/home/<username>/</code>	Local user's home/FTP directory
<code>/var/log/</code>	System and service logs

Path	Purpose
<code>/usr/sbin/vsftpd</code>	FTP server executable

14.2 Important Ports

Port	Purpose
TCP 21	FTP control connection
TCP 20	Traditional Active FTP data connection
Passive port range	Passive FTP data connections

14.3 Essential Commands

```

sudo systemctl status vsftpd
sudo systemctl restart vsftpd
sudo ss -lntp | grep :21
sudo journalctl -u vsftpd

```

FTP Client

```
ftp username@SERVER_IP
```

Common commands:

```

ls      → List files
pwd     → Show directory
cd      → Change directory
get     → Download
put     → Upload
bye     → Exit

```

Remember:

`/etc/vsftpd.conf` → Configuration

`/home/<username>/` → User files

TCP 21 → Control

Active → Server initiates data

Passive → Client initiates data